

Experiment 10

1. Experiment Title

Load and Display JSON Data Using `fetch()` and jQuery `$.getJSON()`

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript
- jQuery
- Live Server Extension

5. Theory

5.1 JSON

JSON (JavaScript Object Notation) is a lightweight format used for storing and exchanging structured data. It represents data using objects and arrays.

JSON is commonly used when exchanging data between a web application and a server.

5.2 `fetch()`

The JavaScript `fetch()` function is used to request data from a resource such as a JSON file or API.

The `response.json()` method converts the JSON response into JavaScript data that can be processed by the program.

5.3 jQuery `$.getJSON()`

jQuery provides the `$.getJSON()` method for loading JSON data.

It provides a simple way to request and process JSON data.

5.4 Dynamic Table Rendering

After loading the JSON data, JavaScript uses `forEach()` to process each object and creates table rows dynamically.

```
data.forEach(student => {  
  
    const row = document.createElement("tr");  
  
    row.innerHTML = `  
        <td>${student.id}</td>  
        <td>${student.name}</td>  
        <td>${student.email}</td>  
    `;  
  
    tableBody.appendChild(row);  
  
});
```

Thus, the table is generated automatically from the JSON data rather than entering every row manually.

4. Output

Dynamic Data Board

Below is the data rendered inside an asynchronously managed UI engine.

Kashish Gupta 24070521235

Update via Fetch APIUpdate via jQuery

Showing 10 baseline records. Use buttons to reload from the live server API.

ID	Employee Name	Email Destination	Assigned Business
101	Kashish Gupta	KashishGupta@example.com	Nagpur Tech Solutions
102	Priya Patel	priya@example.com	Indore Cloud Labs
103	Rohan Das	rohan@example.com	Mumbai Digital Corp
104	Ananya Iyer	ananya@example.com	Bengaluru AI Systems
105	Kabir Mehta	kabir@example.com	Delhi Cyber Logistics
106	Sneha Reddy	sneha@example.com	Hyderabad FinTech
107	Amit Joshi	amit@example.com	Pune AutoSystems
108	Divya Nair	divya@example.com	Kerala EduTech
109	Vikram Singh	vikram@example.com	Rajasthan Solar Grid
110	Nisha Verma	nisha@example.com	Kolkata Data Analytics

5. Result / Conclusion

The experiment was successfully implemented using HTML, CSS, JavaScript, JSON, and jQuery. JSON data was loaded using both the `fetch()` API and jQuery `$.getJSON()`, and the retrieved records were dynamically rendered into an HTML table. This experiment demonstrated how web applications can retrieve, process, and display structured JSON data dynamically.

Thus, the experiment successfully demonstrates loading and displaying JSON data using `fetch()` and `$.getJSON()`.